11 f and g are functions such that

$$f(x) = \frac{2}{x^2} \quad \text{and} \quad g(x) = 4x^3$$

(a) Find f(–5)

$f(-5) = \dfrac{2}{(-5)^2} = \dfrac{2}{25} = 0.08$

1 mark
Square the whole negative input

0.08

(1)

(b) Find fg(1)

fg(1) means f(g(1)): apply g first.

$g(1) = 4 \times 1^3 = 4$

1 mark
Evaluate the inner function

$f(4) = \dfrac{2}{4^2} = \dfrac{2}{16} = \dfrac{1}{8}$

1 mark
Evaluate the outer function

$\dfrac{1}{8}$

(2)

(Total for Question 11 is 3 marks)